Cyber-Attack Type Prediction Using Synthetic Data and Assessing the Impact of Human Factors on Critical Infrastructure Systems

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Abstract

Critical infrastructure systems can cause huge impact if they were a target for a cyberattack, and a lot of studies discuss the risk but there is not an evaluation of machine learning models using a dataset or discussing the role of human factors. This study will evaluate machine learning models such as Random Forest, Support Vector Machine, XGBoost, K-Nearest Neighbors, and Naïve Bayes using a synthetic secondary data that comes from the Global Cybersecurity Threats (2015 - 2024). Also, we will analyze the human factors effects that will be analyzed from the primary data that will come from a questionnaire. The results showed that all the models had a very weak performance with accuracy rates between 15% and 18.6% and Random Forest had the highest performance. It also showed that training, and insiders' recognition are critical for achieving critical infrastructure systems protection. These findings show that predicting attack types using machine learning models on synthetic datasets is challenging because of factors like the data quality.

Keywords: Cybersecurity, Critical Infrastructure Systems, Attack Type Detection, Global Cybersecurity Threats, Random Forest.

1. Introduction

In these days, most of the people use internet in their daily lives, and the use of internet may be dangerous as for example the bad use of internet could lead to an unauthorized access through phones, computers, systems, or even a critical infrastructure system which could lead to unauthorized accessing, editing, or losing data. So, we need cyber security because cyber security is about protecting information and machines such as hardware, software, computers, networks, data, phones, vulnerabilities that could be exploited by cyber criminals, and critical infrastructures systems from losing the availability, integrity, or the confidentiality of these information and machines, as cyber security helps by protecting systems against unauthorized access, editing, or losing these information [1].

Critical infrastructure systems are systems that appear in many sectors such as minerals, health, energy, transportation, and telecommunications and these critical infrastructure systems operate as the basis for an organization, nation, or even a national economy and security, as they act as a backbone for communities and functioning the nations [2]. These systems' failures have a huge impact on the economy and security, so, we need to protect these systems by addressing attacks, vulnerabilities, threats, defensive approaches, and risk plan, assessment, and mitigation [2].

In this section, we will explore the questions and the objectives for this research, but before we discuss the research question, we must understand our problem which is securing critical infrastructure from cyberattacks. We analyzed previous research papers to build a new question for this research.

Research Questions:

- RQ1: How well can AI models such as Random Forest, Support Vector Machine (SVM), XGBoost, K-Nearest Neighbors, and Naive Bayes identify and detect cyberattack types on critical infrastructure systems using synthetic historical data from the dataset Global Cybersecurity Threats (2015 - 2024)?
- RQ2: How human factors like training, and insiders affect the protection of critical infrastructure systems from cyber-attacks?

This paper will show an analysis of cyberattack types on critical infrastructure systems and how AI models can identify these types. Also, we will show a list of human factors that can affect the overall protection of critical infrastructure systems such as training, and insiders threats.

Section 2 presents the related work. Section 3 discusses the data collection and description. Section 4 illustrates the proposed approach. Section 5 presents the experiments and results. Finally, Section 6 concludes the work and provides recommendation based on the conclusion.

2. Related Work

There are a lot of studies that studied the machine learning role in critical infrastructure systems, and the critical infrastructure systems protection including threats and risks associated with it. In this section, we will review these studies focusing on there objectives and results.

Oliveira et al. provide a sequential approach and evaluate multiple machine learning and deep learning on the CIDDS-001 dataset for intrusion detection in single-flow and multi-flow techniques, and the paper was developed for Air Transport Infrastructures of Europe project [3]. They used the Coburg Intrusion Detection Data Set (CIDDS-001) which contains network traffic data that is collected from different environments, and it may be normal traffic or dangerous traffic with many attack types [3]. Oliveira et al. used machine learning and deep learning models, as they used Random Forest, Multi-Layer Perceptron, and Long-Short Term Memory. Single-flow results show that there isn't a way to choose the best model because all models accuracy and other evaluation metrics were very nearby [3]. Multi-flow results show that LSTM performed better than the other models with less-false positive errors [3].

Obuse et al. explore artificial intelligence with security orchestration, automation, accuracy and response (SOAR) to improve the management of incidents within critical infrastructure systems [4]. They did not mention the dataset name, but they said they applied Artificial intelligence models by using datasets that contains historical cyber incidents records and available threat intelligence feed to the public [4]. Obuse et al. used machine learning, and deep learning along with the integration of SOAR, these models are machine learning models through supervised and unsupervised learning, and deep learning models such as Convolutional Neural Networks, recurrent neural networks [4]. The results show that the use of AI withing SOAR improve incident response efficiency, as it reduced the mean time to detect and mean time to respond [4].

Bushigampala and Inaganti provided an AI framework that focuses on detecting threats and response in critical infrastructure systems such as energy grids, water treatment, and transportation networks [5]. They did not mention a specific dataset name, but they collected traffic data from Industrial Control System through sensor-level acquisition that includes protocol, payload, cyclic control commands, and timing patterns [5]. Bushigampala and Inaganti used a hybrid deep learning model that is a combination between C Convolutional Neural Networks and Long Short-Term Memory [5]. The results

show that the hybrid model showed a better performance compared to traditional intrusion detection systems especially in false positives as it achieved a 93% accuracy rate [5].

George et al. shows threats of cyberattacks that are growing in multiple sectors such as energy, retail, healthcare, finance, and manufacturing, and it provides an assessment of cyber vulnerabilities across these sectors [6]. They did not mention a dataset, but they collected their data from government, association industry, finance, academic journals, audit findings, and news reports [6]. George et al. did not use machine learning models, instead, they relied on five evaluation criteria such as financial loss exposure, sensitivity of compromised data, reliability and safety impact, cyber readiness, and regulatory requirements [6]. The results show that there are challenges withing the sectors such as legacy systems, skills gaps, and weak security by design practices [6]. Also, they said there are different threats for each sector such as stealing medical records, payment platforms damage, etc. that are exploited through ransomware, and website attacks [6].

Leroy and Zolotaryova provide and exploration of the components of cyber autonomy to defend against cyber attacks within critical infrastructure systems based on experts point of views [7]. They did not use a specific dataset, as the study collected data from 8 experts from the European Union and United States of America that have over 15 years of experience in critical infrastructure and have an economic and technical background [7]. Leroy and Zolotaryova did not use AI models or conducted a machine learning experiment, but they used the delphi method using interviews, experts assigned weights as a scale from 1 to 5 [7]. The results showed that the most critical elements for achieving effective cyber autonomy within an organization are policies, reputation management, infrastructure and architecture [7].

The studies provided a good contribution and knowledge in securing critical infrastructure systems, but there are some gaps to these studies. As some papers only focus on the technical aspects of detecting attack types on critical infrastructure systems, other papers rely on expert opinions or frameworks without a real use of dataset along with AI models. So, there is a lack of studies that combine technical and human aspects in critical infrastructure systems which motivated us to produce a research paper that combines technical and human aspects in critical infrastructure systems and use a dataset along with AI models that will help us do experimental analysis to support our results and solutions.

3. Data collection and Description

Primary data are those data that did not exist before and it is the responsibility of the researcher to produce these data. The secondary data differs from the primary data as the secondary data are those data that exist before and the researchers rely on these data to make his research just like the datasets on Kaggle, a key element is that secondary data generation rely on primary data as in order to generate secondary data, a primary data must be generated.

3.1 Primary Data

The objective of the questionnaire is to explore how human factors (training and insider threats) impact the success of securing the critical infrastructure systems from cyberattacks which meet the RQ2: How human factors like training, and insiders influence the success of the security of the critical infrastructure systems? So, we aim to collect primary data with a questionnaire that has only closed-ended questions that will be listed below:

Question	Why it was chosen
Q1: What is your role as an IT student?	The target audience are only IT students, so this question was written to make sure that the audience responses are part of the target audience.
Q2: How familiar are you with cyber threats that target critical infrastructure systems	As RQ2 sees the human factors and one of them is awareness, this question aims to understand the awareness of the audience about cybersecurity threats that target the systems. (Awareness)
Q3: How familiar are you with different types of cyber threats (e.g., malware, phishing, worms).	This question is also to measure the awareness and knowledge of different types of cyber threats. (Awareness)
Q4: How often do you explore updates on new cyber threats?	As RQ2 sees human factors and one of them is training so this question was chosen to see how often IT students update their knowledge on cyber threats. (Training)
Q5: Do you believe that cybersecurity training is important to achieve critical infrastructure systems security?	The question is also about training, and it aims to see the audience's opinions about training if they improve the security of systems. (Training)
Q6: Do you agree that insider threats are one of the main risks on critical infrastructure systems?	As RQ2 sees the human factors and one of them is insider threats, this question aims to determine if the people are aware that insider threats are one of the main risks. (Insider threats)
Q7: How confident are you that organizations can detect and prevent insider threats?	This question is also about insider threats so, this question aims to see the audience's opinions about organization's ability to prevent and detect insider threats. (Insider threats)
Q8: How important do you think human factors (e.g., training, insider threats) are	This question is general and aims to see the audience's point of view about the

for protecting critical infrastructure systems?	importance of human factors on protecting systems. (Human Factors)

Table 1: Questionnaire

The process started with selecting questions that serves the goal of RQ2, as we put in questions that helps us answer the RQ2, then we created the survey with Google Forms, and it has been sent only to IT students from different universities in Jordan.

We had so many difficulties trying to get responses, as it was very hard to get a good number of responses, as we sent the survey to a lot of students, and we sent it and reminded them multiple times. Finding the target audience was very easy but getting responses from the target audience was very difficult and it delayed the whole data collection process.

There are multiple merits and limits for the questionnaire we made:

- The merits:

    1. Data analysis did not take much time, as google forms provide a data analytics dashboard.
    2. Currently most IT students have knowledge in cybersecurity, even IT students that just arrived at university so all IT students can be able to answer questions.
    3. The survey included different human factors to assess which help us fully answer the research question.

- The limits:

    1. Responses may not be fully correct because they provide personal knowledge or experience not data that is based on a very trusted audience.
    2. The collection of this data may take a lot of time as we want only the target audience to answer, fill, and complete the questionnaire, not anyone else.

3.2 Secondary Data

The dataset Global Cybersecurity Threats (2015 – 2024) is a synthetic dataset that shows data on cyberattacks, malware types, targeted industries and affected countries and it contains more than 3000 rows and only 10 columns, 6 columns data type is String, 3 is integer, and 1 is decimal. It was created to track these cybersecurity incidents, threats, and attack vectors to help with threat analysis, cybersecurity trend forecasting, and machine learning model development for one goal which is enhance global digital security [8].

The dataset was generated based on some factors, as it proposes a simulation of global cybersecurity Threats from 2015 to 2024. The dataset columns were generated based on factors such as commonly reported metrics in cybersecurity, historical reports, expert written papers, and publicly available reports and frameworks. Also, they mentioned that the dataset should only be used for academic or professional context, as it has a synthetic nature [8].

Column	Description
Country	The Country where the attack happened.
Year	The year when the incident happened.
Attack Type	The Class Label, which shows the type of attack such as Malware, DDos, etc.
Target Industry	The targeted industry from the attack such as Finance, Retail, Government, etc.
Financial Loss	Estimated Financial loss in millions.
Number of Affected Users	The number of impacted users because of the attack.
Attack Source	It shows the origin of the attack.
Security Vulnerability Type	The type of vulnerability in the industry that has gotten exploited
Defense Mechanism	The cyber defense mechanism used such as Firewall, IDS, IPS, etc.
Incident Resolution	The full time taken to resolve the incident.

Table 2: Dataset Columns

There are many merits of the dataset, for example, it covers a wide range of cybersecurity threats on many industries, and it shows the affected countries. Also, it covers the cybersecurity threats over many years which is ten years. On the other hand, there are limitations of the dataset such as, the data is not real-world data, but it is a simulation that is generated. Also, it has not been verified by cybersecurity professionals or firms, and it is only intended for machine learning, data visualization, storytelling, trend and pattern analysis, and cybersecurity education.

I choose the dataset because it covers everything I needed and covers it not only for three or five years, but for ten years as I needed something that aligns with my research which is securing critical infrastructure and the dataset helps me to identify and understand such attacks, defensive mechanisms, and attack sources and their consequences such as financial, national, and global loss.

4. Research Approach and Methodologies

In this section, we will explore the onion model and provide the terms that are followed in this research, and we will provide the philosophy, theory development approach, methodological choice, research strategies, time horizons, and techniques and procedures. Also, we will provide the research methodology in terms of the process done on the secondary data.

4.1 Onion Model

The onion model is a model that contains 7 layers, and each layer helps students to produce an organized methodology. The onion model was presented in a book called "Research Methods for Business Students" which clearly shows that the model is designed for students, and it was presented by Mark NK Saunders, P.Lewis, and A.Thornhill.

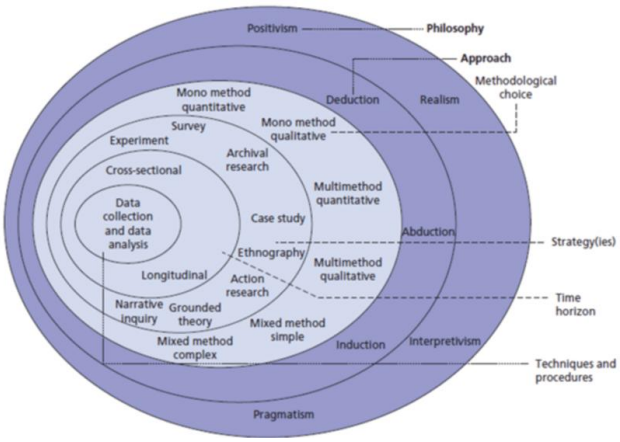


Figure 1:The structure of the Onion Research Model [9].

4.1.1 Philosophy

The chosen philosophy is critical realism, because critical realism shows that reality exists like attacks, attack types, and vulnerabilities but humans have different points of view based on their experience. Also, these attacks exist, but human factors like training, and insiders can affect these threats in many ways like how these threats are managed. Critical realism shows the technical part in RQ1 and the human point of view in RQ2. Critical realism is suitable for the study because the study assumes that cyber security threats exist independently of human perception, and their detection and management is influenced by human factors.

4.1.2 Theory Development Approach

The chosen development approach is Inductive, because induction means I observe something in the data which and then generalizes it to form a conclusion which fits my research. in my research the conclusions are created from the data as the AI model will identify types of cyberattacks on critical infrastructure systems from the dataset, and the questionnaire shows how human factors can affect the protection of critical infrastructure systems, as I create conclusions without relying on existing theory. We

did not select deductive because the study does not text an existing hypothesis, as we want to do experiments and figure conclusion independently.

4.1.3 Methodological Choice

There are huge differences between quantitative and qualitative research, as quantitative research relies on data that can be represented numerically data like counts, percentages, and scales and most of the time it uses closed-ended questions. Qualitative research relies on data that is represented in non-numerical data like words, and opinions and most of the time it uses open-ended questions.
The chosen methodological is Mono Method Quantitative, because RQ1 and RQ2 rely on numerical analysis. The dataset for RQ1 has categorical values such as attack type, and industry are analyzed quantitatively using accuracy, precision, recall, and f1-score and can be preprocessed by encoding. The questionnaire for RQ2 uses scale responses from 1 to 5 which allow quantitative analysis, along with only quantitative questions.

4.1.4 Research Strategy

The research strategy chosen is a combination between survey and experiment because for RQ1 we will do an experiment on the algorithm to measure the performance based on some evaluation metrics. For RQ2 a questionnaire is used that is answered in a scale way from 1 to 5 to measure human factors effect on critical infrastructure protection. This combination between survey and experiment strategies helps us to evaluate technical aspects along with human factors aspects.

4.1.5 Time Horizons

The chosen time horizon is Cross-Sectional, because the process of collecting and analyzing data is done at one time, the dataset is already completed and my responsibility only to collect the data in it and analyze it to identify attack types at one time, the questionnaire also is created and collect data from participant at one time. We did not choose longitudinal because the study does not do the process on changes over time but analyzing everything at a specific point.

4.1.6 Techniques and Procedures

There are different techniques and procedures used in my research. For RQ2, I used google forms to create the questionnaire, because it's simple and easy to create questionnaire with, I can create a questionnaire very fast and responses can be exported in different formats, and google forms helps me to make an analysis as it shows different type of charts which makes it easier to measure awareness, training, and insiders affection on critical infrastructure systems. For RQ1, I used python in preprocessing model training, and evaluation, as we did the preprocessing steps, the modeling training process, and the evaluation process using python.

4.2 Research Methodology

In this section we will explain the process of answering the research question, including the process that has been done on the secondary data, the models used in these secondary data, and the metrices of the models for the secondary data.



Figure 2:The structure of the Research Approach.

4.2.1 Preprocessing

First, we performed preprocess for the secondary data to make the dataset fully prepared for the modelling and analysis to produce a high-quality output. Preprocessing was essential step to make sure that the data is ready for machine learning models.

1. Removing Columns: We first looked at what columns should be removed from the dataset, and it was clear that no columns should be removed, even though there are some columns that have a lot of unique values such as "Number of Affected Users", but they are still lower than the total number of rows of the dataset.

2. Handling Missing Values: Also, there were no missing or null values in the dataset, so we did not remove any row, and fill any missing values because we don't have any missing values.

3. Label Encoding: We have a lot of categorial important columns, but we must convert them to numerical columns because AI models require numerical columns, so we have categorical columns such as "Country, Attack Type, Target Industry, Attack Source, Security Vulnerability Type, Defense Mechanism Used", these columns were converted to a numerical columns using label encoding which is a way to convert the category into a unique number to make it suitable for AI models.

4. Components Dividing: We divided the dataset into two components, X for the features which include all the columns but not the class label, and y for the target which is the class label "Attack Type".

5. Standardization: Lastly, we made standardization of numerical columns, and we did it specifically for one reason, because we have in the data set high numbers and low numbers we don't want make the AI model thinks that higher numbers are more important than lower number, so the standardization means making the standard deviation = 1 and the mean = 0 which makes all the numbers are almost near each other.

4.2.2 Modeling

In this section, we will perform the modelling process using different models such as Random Forest, SVM, XGBoost, K-Nearest Neighbors, and Naive Bayes. As we divided the dataset for testing and training, 20% of the data were for testing, and 80% of the data was for training.

1. Random Forest: Random Forest was used in modelling, and it is a machine learning tool that works by building decision trees using sections of the training data and random subsets of the features at each split of a tree. Each tree has a prediction, and the final classification prediction is calculated by majority voting which helps reduce overfitting and handle weak features [10].

2. SVM (Support Vector Machine): SVM was used in modelling, and it is a learning method that is used for classification and regression, and it works by finding the best separating boundary, and the goal of SVM is to maximize the distance between the boundaries and the nearest points from each class, where these points are called support vectors [11].

3. XGBoost was also used in modelling, and it is an algorithm that works by gradient boosting where multiple small decision trees are built and each tree correct the errors made by the previous trees and it uses multiple CPU cores to organize the data and reduce training time [12].

4. K-Nearest Neighbors: K-Nearest Neighbors was also used in modelling, and it works by looking at the nearest data points to a given row using a distance measures and finding the most similar ones and the prediction is made based on the neighbors that are closest to the row [13].

5. Naive Bayes: Naive Bayes was also used in modelling, and it is an algorithm that uses probability classification based on bayes theorem "a theorem that identify conditional probability of event A when event B has already happened [14]", the algorithm calculates the probability of each class for a row and then it selects the highest probability class [15].

More than one machine learning model is applied in this research to see the performance of each one, and we selected these algorithms because each one of them has their different strengths which provides a clear understanding of the dataset and allows us to compare performance.

4.2.3 Evaluation Metrics

For the modelling algorithms, we showed the outputs and the results based on specific metrics and criteria, and they were Accuracy, Precision, Recall, and F1-Score.

1. Accuracy: Accuracy means how accurate the algorithm is and how many correct classifications were made across the test, so if the accuracy rate is 70% then the error rate is 30%, as it only shows the overall performance [16]. We use precision, Recall, and F1-Score along with accuracy because sometimes the accuracy does not work alone.

2. Precision: Precision means how much the algorithm was accurate in the results, at it is the ratio of true positive records across all the predicted positives records which make sure about what really was true in the positive records which also makes us understand what are the false positive records in the positive records, for example an algorithm had 10 positive records but only 8 were truly positive so there are 2 that are false positive [16].

3. Recall: Recall means looking over the true positive rate, but it is different than precision, as for example, we have 6 attack types in the dataset, what is the ratio of the number of attacks that the algorithm concluded, it actually see the percentage of true positive records, for example an algorithm had 10 positive records containing only 4 attack types while the dataset have 6 attack types, so there is 2 attack types that are false negative and the model did not see the two attack types it only saw 4 attack types [16].

4. F1-Score: F1-Score is measure that combines precision and recall measures, basically, it is the average of precision and recall for example the precision is 18% and recall is 16%, the F1-Score will be 17% so the F1-Score calculated by 2 * (precision * recall) / (precision + recall) [16].

More than one evaluation metric is used to give a more accurate, reliable, and understandable evaluation and allow us to compare different algorithms based on different evaluation metrics, and higher number indicate better performance for the four evaluation metrics.

5 Results and Discussion

In this section we will provide the results of the primary and secondary data and answer the research questions based on the results, as we will provide the results of the primary and secondary data and discuss them in order to answer the research questions.

5.1.1 Secondary data results

In this section, we will present and discuss the performance of machine learning models for classifying attack type on critical infrastructure systems.

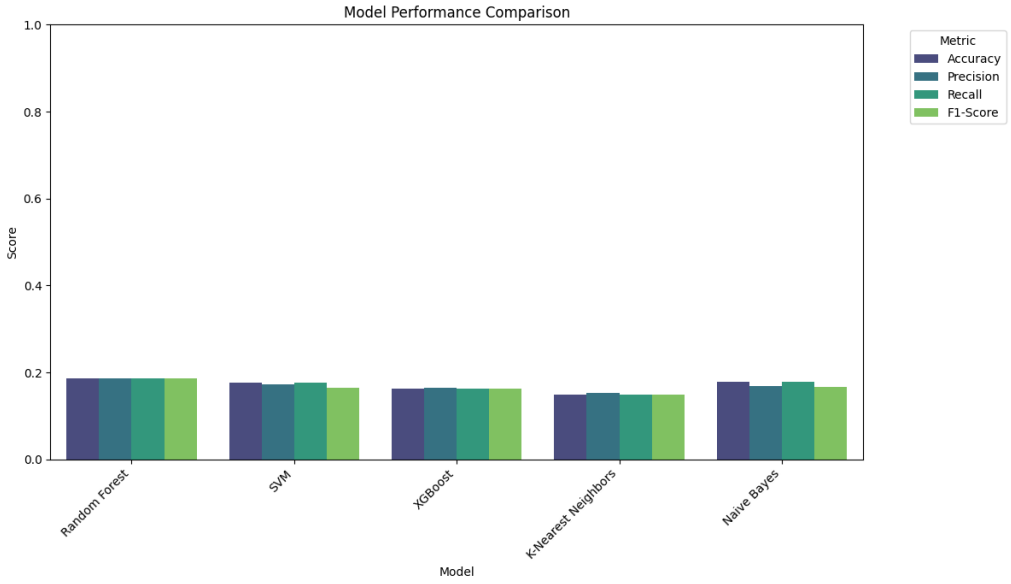


Figure 3: Model Performance on Predicting Attack Type.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	**0.186667**	**0.185907**	**0.186667**	**0.185689**
SVM	0.176667	0.172799	0.176667	0.165654
XGBoost	0.163333	0.164823	0.163333	0.163686
K-Nearest Neighbors	0.150000	0.153553	0.150000	0.148710
Naïve Bayes	0.178333	0.169568	0.178333	0.165796

Table 3: Model Performance on Predicting Attack Type.

Table 3 summarizes the classification performance of the evaluated models. The performance of all models was very low, and all models' performance was very weak, but even though Random Forest achieved the best overall performance among all the bad model performance with an accuracy of 18.66%, precision of 18.59%, recall of 18.66%, and F1-score of 18.56% which is the better one among all classifiers. Which indicates that, among all the models, the higher one has a very low percentage of classifiers which shows that the performance of all the models used is very weak.

Naïve Bayes ranked second because it has an accuracy of 17.83% and an F1-score of 16.57%, even though SVM was very nearly to be ranked second. Which shows that most of the percentages on classifiers between models were near to each other and they all were considered very weak. But, XGBoost model had a moderate performance compared to the performance of the other models.

Even though all models performed weak, we have the weakest performance, which is K-Nearest-Neighbors with an accuracy of 15% and F1-score of 14.8%.

All the results show that the best model accuracy is 18.66% and f1-score 18.5% but the weakest model accuracy is 15% and f1-score 14.8% which indicates that all the models were very bad and in comparison, there is only 1% to 2% difference between the models.

None of the related work used the same dataset or provided an actual model performance that can be compared to this research which shows that this study fills the gap in the previous ones by using machine learning algorithms to a cybersecurity dataset.

Overall the results of the machine learning models show that the five models had a weak performance with accuracy values between 15% to 18.66% and with an f1-score values between 18.56% to 14.8% which shows that the algorithm perform weak on a dataset like the Global Cybersecurity Threats (2015 - 2024) which may have been because the dataset does not have strong features that makes machine learning models learn from them.

The AI models could not detect cyberattack types on critical infrastructure systems effectively and all the percentages were below 19% for all the models which shows that these models are not reliable to predict the attack type on critical infrastructure systems using the Global Cybersecurity Threats (2015 - 2024) dataset as a synthetic dataset. Also, it indicates that the dataset may does not have all the information needed or it does not have strong information for predictive modeling. So, the result is that AI models such as Random Forest, Support Vector Machine (SVM), XGBoost, K-Nearest Neighbors, and Naive Bayes cannot effectively identify cyberattack types on critical infrastructure systems using synthetic data due to a lot of factors.

There are limitations and factors on that study that affected the overall performance of the models such as that the dataset used was synthetic and does not show real-world cyberattacks but rather than that it is simulated. Also, the dataset did not include technical columns such as the network and packet flow which made the predictions weaker. Also, the classification and preprocessing for the dataset could be weak, as it may need stronger classification and preprocessing techniques to enhance the prediction performance process. Overall, the study showed that attack type prediction on critical infrastructure systems using synthetic data is a very challenging problem.

In my opinion, the future direction should be extending the study using a real-world cyber-security dataset that provide real incidents and patterns with more detailed technical features such as the network traffic and packet flow which could raise the performance of the predictive models, give a reliable result, and give more realistic results.

Also, future work could include applying more advanced machine learning models or even deep learning models, using advanced preprocessing techniques, and comparing the predictive results from using synthetic dataset and real dataset to see a the real effects of data quality.

5.1.2 Survey results

In this section, we will show the survey that was made to explore how human factors like training, and insider threats influence the protection of critical infrastructure systems.

The survey support research question 2 and it was designed especially for IT students from different academic years as we sent the survey to the IT students at Al-hussein technical university and to IT students from different universities, as the target audience was only IT students which indicates that all responses come from people that have at least basic knowledge to cybersecurity.

The survey included 8 closed-ended questions, one question for overall opinions, two questions for each human factor like training, and insider threats, and one question for specifying the respondent role.

After sending the survey to IT students from all over Jordan, we collected 37 responses that will help us to answer research question 2.

The survey had only three major core questions that will answer research question 2 and there are other supporting questions, as the survey results show that all of the respondents are IT students from different majors, as 40.5% were computer science

students, 29.7% were cybersecurity students, 27% were AI students and the others were from other majors.

Most respondents were familiar with types of cyberthreats like malware or phishing, as 86.4% of the respondents show a moderate to very high familiarity with different types of cyberthreats, and only 13.5% of the respondents show a very low familiarity or zero familiarity.

Most respondents have a moderate to high awareness of cybersecurity threats that target critical infrastructure systems which is a strong indicator of the awareness of the respondents as most of respondents have a moderate to high awareness of cybersecurity threats that target critical infrastructure systems, and only a small group of respondents have very low to low awareness of cybersecurity threats that target critical infrastructure systems which shows that the survey answers for other question will be strongly reliable as most of the respondents have a good knowledge and awareness.

The survey shows moderate to very high results to the training part, as the training was considered as exploring updates on new cyber threats, and the survey shows that more than 50% of the respondents explore updates on new cyber threats frequently and regularly but there are a few people who always explore updates, and the others sometimes or never explore updates, which indicates that most of the audience explore updates which is an indicator that they are aware because if they explore updates on cyber threats, then they are aware of these cyber threats.

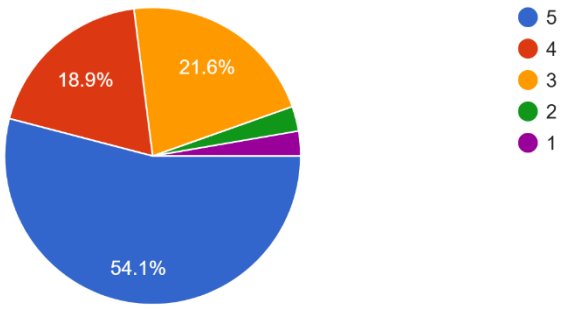


Figure 4: The importance of training to secure critical infrastructure systems (1 = not important at all, 5 = very important).

Figure 4 shows the respondents opinions on the importance of training to achieve the security of critical infrastructure systems, as most of the respondents agree or strongly agree that the training is very important and a critical part of securing critical infrastructure systems which indicates that there is a strong understanding of training within the critical infrastructure systems.

The survey also shows the confidence that the respondents have on organizations to detect and prevent insider threats, and the results were strong as a lot of the respondents shows a moderate and high confidence of organizations to detect and prevent insider threats, and a very small group only were not very confidence.

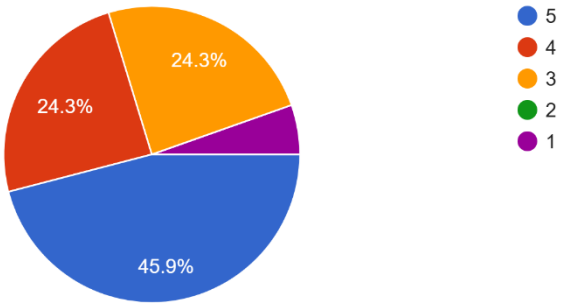


Figure 5: Insider threat as a risk on critical infrastructure systems (1 = strongly disagree, 5 = strongly agree).

Figure 5 shows the scale of the risk that is connected to insider threats on critical infrastructure systems, as the results were very strong according to figure 5, and most of the respondents agree or strongly agree that insider threats are on the main risks on critical infrastructure systems which shows a strong awareness of insider threats.

Overall, based on the results provided in 5.1.2 section and based on the audience answers, we can now say that research questions 2 is answered, as the findings shows that human factors affect the overall protection of the critical infrastructure systems. There is a high level of awareness of cyber threats that target critical infrastructure systems which shows that understanding the cyber threats is the first step to achieving the security of critical infrastructure systems, as it allows employees to recognize the risks. Also, most of the people agreed strongly that the training is important to achieve the security of critical infrastructure systems which shows that trained employees have a critical role in reducing and detecting cyberattacks, for example, employees who are trained on phishing and social engineering attack will not be susceptible to these attack, employees who are trained on incident response plan can apply the plan more effectively, and employees who are trained on different types of attacks that target critical infrastructure systems will make the employees recognize these risks Fastly and raise their awareness. Also, audience recognize that insider threats are one of the important risks within an organization to secure critical infrastructure systems, so organizations should prevent insider threats through training, and clear policies. These factors improve early detection, correct and fast response, and reduce human errors. All the results shows that the protection of critical infrastructure systems does not only depend on technical parts but also it relies on employees training and awareness, and recognition of insider threats.

There are limitations of this study, as we only focused on IT students rather than focusing on experienced employees who directly work in critical infrastructure systems to get more real actual answers rather than students' opinions. The responses size was small, and it needed to be larger to get more reliable insights. Also, we only focused on two human factors while there are other important human factors that can strongly affect the overall protection of critical infrastructure systems.

Future work should be expanding the target audience and collect data from experienced employees such as engineers, cyber security employees, and employees who work inside critical infrastructure systems. Also, we should collect larger responses from different types of audience, use interviews and other data collection methods along with the survey, and assess more human factors.

6 Conclusion and Recommendations

The study aims to evaluate machine learning models to identify cyberattack types and explore the human factors affecting the protection of the critical infrastructure systems.

The findings showed that respondents strongly agreed on the importance of training, indicated that insider threats are major risks to critical infrastructure systems, and strongly agreed on the importance of all human factors to protect critical infrastructure systems.

The findings also showed that the machine learning models' performance was poor and the best model was Random Forest with an accuracy of 18.66% and the weakest model was K-Nearest Neighbors with an accuracy of 15%, so modeling showed that data quality and performance play a crucial role to achieve a strong modeling performance.

In real world situations protecting critical infrastructure systems does not only depend on predictive analysis, but it also depends on employees training, and insider threats recognition, also, predictive modeling strongly relies on data quality.

There are limitations of this study, for secondary data, we used a synthetic dataset that did not have any technical features. Also, the responses were small due to multiple factors such as that the responses were targeted only for IT students and it should have been to IT students and anyone with IT background to raise the reliability of the responses.

Future work should be replacing synthetic dataset with real-world dataset that include technical features, applying more advanced machine learning models, advanced preprocessing techniques, and comparing synthetic results with real-world results. Also, future work could include expanding the target audience, collecting larger responses, assessing more human factors, and conducting interviews with surveys.

Based on the findings of primary data, there are recommendations that can be proposed to the party responsible for protecting critical infrastructure systems.

Because most respondents show a moderate to high awareness of cyber threats, only a small group of the respondents explore updates on new cyber security threats, others do it sometimes or rarely, and the majority of the respondents strongly agree that cyber security training is essential and critical aspect to secure critical infrastructure systems, which shows a gap in continuous cybersecurity learning, so, we recommend organizations responsible for securing critical infrastructure systems to implement regular training programs every 3 months and focuses on current and new cyber threats, attack scenarios, awareness on social engineering attacks, and how to apply the incident response plan.

This recommendation would improve the employees awareness of new and existing cyber threats which leads to the recognition of attacks if they occur, reduce human errors by raising the awareness level of social engineering and phishing attacks, improve incident response plan understanding among the employees which leads to faster application of incident response plan procedures, and overall reduce the attacks which reduces the financial loss and damaged systems.

This recommendation has limitations such as that training programs could be costly and require time and more resources, but none of that can be compared to the financial loss if an attack happens, and training alone cannot fully prevent attacks, but the organization also needs strong technical controls.

Also, because most respondents agree or strongly agree that insider threats are on of the main critical risks to critical infrastructure systems, and showed that they have a moderate to low confidence in organization to detect and prevent insider threats which shows that

existing controls are not effective and need more monitoring procedures, so, we recommend organizations responsible for securing critical infrastructure systems to improve the management of insider threats programs by adding clear access control policies, adding clear laws and penalties for insiders, add permissions based on roles, and implement strong technical and human monitoring procedures because these monitoring procedures would help the organization to detect insiders in the least time possible.

This recommendation would reduce unauthorized access to the system by using permissions based on employees' roles, improve early detection of insider threats because of continuous technical and human monitoring, increase the understanding by clearly having policies and consequences for insiders, and raise the organizational trust to secure and protect critical infrastructure systems.

This recommendation has limitations such as that monitoring the employees could raise privacy and ethical issues, implementing the monitoring and access control systems could be costly, and employee productivity may be affected if controls were very strict.

Based on the findings of secondary data, there are recommendations that can be proposed to the party responsible for protecting critical infrastructure systems.

Because the results of the machine learning models showed a weak performance with accuracy values between 15% and 18.66%, and all models failed to differentiate cyberattack types on critical infrastructure systems effectively using synthetic data, which shows that the synthetic datasets lack strong features for accurate cyberattack type prediction, so, we recommend organizations responsible for protecting critical infrastructure systems to do a cyber-attack predictive analysis from time to time but by should also review and improve quality and structure of dataset used for predictive analysis by including technical and operational features such as network traffic, packet flow, and system logs before doing predictive analysis.

This recommendation would improve the accuracy and reliability of cyber-attack type prediction on critical infrastructure systems by using models with useful technical and operational features instead of relying on synthetic data, including network traffic, packet flow, and system logs to understand the real attack scenarios and behaviors which can help organizations to detect attacks early with a high accuracy.

This recommendation also has limitations such as that operational cost may decrease because collecting technical and operational data needs skilled employees, and predictive analysis cannot fully replace existing controls, but it should be a combination between monitoring, detection and prevention tools, and predictive analysis.

Also, because all models showed a very weak and very similar accuracy values, so no single model was reliable and effective enough to predict cyberattack types on critical infrastructure systems, so, we recommend organizations responsible for protecting critical infrastructure systems to not only rely on one model for predictive analysis, use multiple machine and deep learning models for predictive analysis, and use the output as a supporting tool with existing controls and with human expertise rather than a replacement for existing controls.

This recommendation would help organizations to avoid making wrong conclusions and decisions, improve attack detection by comparing results from multiple models, and help security teams by providing early warning indicators.

This recommendation has limitations such as organization may need skilled employees to analyze the results which would increase the employees' cost, and predictive models may give wrong or false predictions that can be noticed by a skilled cybersecurity employees.

7 Reflections

7.1 Selected Research Methodology

The research process started by choosing a field of study and then reading five related papers. After reading the five papers, I noticed that there is a gap in these papers which is not using dataset to assess predictive methods or not even assess human factors on the security of critical infrastructure systems. I decided to explore whether AI models can identify cyber-attack types using historical data and assess how human factors affect the overall protection of critical infrastructure systems.

Understanding the objectives of the research helped guide us in all decisions, as we applied the onion model which helped us a lot in the methodological choices and especially the decision to use a quantitative approach. The quantitative approach helped us to make an effective use of modelling and their performance, and in analyzing the survey responses which was a structured way to answer research questions, but the quantitative method had only one limitation in the survey part, which is we couldn't collect answers from people on why or how some of certain human factors affect the overall protection in their point of view which the qualitative method can support it.

We choose a historical dataset that appeared so good and structured to help us answer the research question 1, but in the process of model training and evaluation, it appears that the dataset have a lack of a real-world scenarios which was not really reliable for predicting cyber-attack types, as we realized that the historical data alone are not reliable for predicting cyberattack types but it must have more technical features and more real-world scenarios.

On one hand Data preprocessing was essential in order to do modelling, as data preprocessing improved data reliability and consistency to make it complete for the AI models. On the other hand, even with doing data preprocessing, the data was not fully complete for AI modelling due to missing or weak features which shows that data preprocessing enhances the data reliability and consistency but cannot solve major dataset problems.

After doing the data preprocessing, we used multiple AI models such as Random Forest, Support Vector Machine, XGBoost, K-Nearest Neighbors, and Naïve Bayes. Using more than one model made the research stronger and made the findings stronger and more reliable, but the weak performance of all the models showed that the limitations was not about the models itself, but it was more about the data we used, as even though the performance was weak but the findings clearly answered research question one.

Also, we used a survey through google forms to assess human factors such as awareness, training and insider threats and it was effective because it helped us to collect structured responses quickly and the form did not take a lot of time from us to be made. The use of quantitative method in google forms helped us to summarize the answers using scales from 1 to 5 and it reduced data mistakes because the respondents did not even have to enter qualitative data which improved data readability. There was a limitation for using surveys which were having difficulties finding the target people to fill the form and the responses size was not very big and we should have used interviews along with the surveys.

We must talk about the tool used for modelling, and it was python, as we did not only use python for model training and evaluation but also for data preprocessing and it was very effective because it helped us to do data preprocessing, training and evaluation across five models which strengthened the findings.

Overall, the research methodology and process were effective in answering the research questions, but it also showed that better datasets, mixed research methods, and better preprocessing techniques were needed to improve the findings.

7.2 Alternative Research Methodologies

Even though the methodologies used were effective but there are alternatives to more effective methodologies that should been used.

We used a quantitative approach, but the alternative would been using a mixed method approach by combining quantitative and qualitative methods. We must say that quantitative was effective in summarizing survey responses, but it did not help us to have a deeper understanding on why and how human factors such as training, and insider threats affect the overall protection of critical infrastructure systems, Also, we could have used interviews or open-ended questions in survey to get stronger answers but the alternative was not used initially because it takes a lot of time, needs professionals, make the analysis process more hard, and make the analysis process takes more time.

We also used the Global Cybersecurity Threats (2015 - 2024) dataset. This dataset was synthetic, lack technical features, and lack of real-world scenarios which make the model performance weak. The use of another dataset could be a good alternative, and the dataset should not be synthetic, have a real-world scenarios, and have technical features like network traffic, and packet trip which would improve the model learning and prediction performance but this alternative was not used initially because we did not realize that these features could affect the performance of the models as we thought the chosen dataset was perfect for modeling purposes.

We also used five machine learning models such as Random Forest, Random Forest, Support Vector Machine, XGBoost, K-Nearest Neighbors, and Naïve Bayes for predicting attack types and they were effective but had a weak performance due to data problems but we could have used more advanced machine learning models or deep learning models that may can deal with the data even if it have problems and these alternative could be neural networks as these model can deal with more complex and data than the chosen models. These models were not used initially because we wanted to first try evaluating the traditional machine learning models, and more advanced models need a very large, high-quality dataset.

For the preprocessing techniques there were alternatives that should have been done to the data such as feature engineering as these techniques might improved the data quality to make a better modelling performance but they were not used initially because we though our preprocessing steps were enough for the data and none of the preprocessing techniques could solve a major problem of the data.

After this journey, I learned a lot of lessons and the main one was that the quality of the data has a high impact on model performance more than the choice of the models themselves, as very good models cannot perform well when the quality of the data is weak.

Other lessons learned that historical and synthetic datasets are challenging in predicting cyber-attack types as good prediction require technical features and real-world scenarios

not simulated scenarios. Also, in every research when dealing with evaluation machine learning models, there should be an evaluation of at least three models to make the findings stronger.

Also, data preprocessing alone can only solve minor data problems but cannot solve major data problems like adding new features.

7.3 Recommended Actions and Future Considerations

The experimental results of models showed that all machine learning models used had a weak performance with an accuracy value between 15% and 18.66% to predict attack type, and this weak performance did not happen randomly but there are reasons mainly a weakness in the dataset.

Random Forest achieved the highest performance between the other used models because random forest can deal with weak features better than other models, and since the dataset contained many categorical columns that were encoded numerically, Random Forest was able to achieve the highest performance by seeing the simple patterns, but other models could not. Naïve Bayes ranked second because it relies on probabilities rather than relationships between features, and since the dataset had weak features, Naïve Bayes was able to make predictions based on probabilities of attack types. K-Nearest Neighbors had the worst performance between the used models because it is a distance-based algorithm, as it compares rows based on the similarity but in the data set the features did not represent real distances which made the similarity calculations unreliable.

All of the results were weak due to data limitations, as the data did not contain strong features, and technical features like packet trip. Also, it lacked real-world scenarios, and the dataset is synthetic and simulated which made the performance of the models very weak. The weak performance of the models does not only depend on the models themselves but also depend on the data used and improving the data is more important than improving the models.

First, we must use another dataset that contains more technical features such as network information, packet trip, protocols used, etc. as these features would help models to understand how attacks behave before they happen instead of relying on historical and general data. Second, we may use a larger dataset, because these models learn better when there is a large data to be trained on to have a better and accurate recognition of attack types of patterns. Third, the chosen dataset was synthetic, simulated, and lacked real-world scenarios, so we must use a real-world dataset that supports the prediction process to produce a real-world solution we must use a real-world dataset that contains real-world scenario, so overall, we must use larger dataset that have technical features, and must not be synthetic or simulated, but must have real-world scenarios to make the findings more realistic..

Fourth, we did effective preprocessing techniques, but better preprocessing can be used to improve performance such as creating new features, and more advanced encoding techniques for categorical data.

Fifth, we could use more advanced machine learning models or even deep learning models to improve performance such as neural networks or even deep learning models and compare the results from using machine learning and deep learning models with synthetic and real-world datasets.

Also, this study could be extended by using the same dataset but testing more machines and deep learning models and comparing their performance, and then we could use the same models but on different datasets and compare their performance.

For the primary data results, it was proved that training and insider threats strongly affect the overall protection of critical infrastructure systems, as training helps employees to detect and sometimes prevent cyber-attacks, and insiders could be a source of some attacks.

First, we must expand the target audience to collect data from experienced employees such as engineers, cyber security employees, and employees who work inside critical infrastructure systems.

Second, we could use mixed research approach instead of quantitative research approach to have a combination of quantitative and qualitative methods and use interviews along with the surveys to collect larger and better answers which would give a deeper explanation on why and how human factors affect overall protection not only measuring their impact.

Also, we could study and assess more human factors other than only training and insider threats because there are a lot of other human factors that can affect the protection of critical infrastructure systems.

Overall, there are multiple future considerations that would help in expanding the study and give more understanding of how well AI models can predict attack type on critical infrastructure systems and what more human factors can affect the protection of critical infrastructure systems.

7.4 Recommended Methodology

The decisions evolved during the whole research process from the initial to the last phase, as these decisions were conducted based on the overall objectives of the study.

At the start of the project, the first decision was choosing a cybersecurity topic, and we chose critical infrastructure systems protection because there aren't many practical studies on these systems, even though these system has a strong influence on society, community, economy, nation, and even on the international aspect. So, we looked at previous studies related to critical infrastructure systems protection and noticed that most of the studies did not focus on experimental analysis, but they focused on experts' opinions or existing case studies and there was limited use of datasets and experiments, and did not asses human factors effects on critical infrastructure systems even though they have a strong effects. As a result, this influenced us to decide that the study should include a machine learning evaluation on critical infrastructure systems and studying human factors effects which lad to producing and developing research question one and two. These research questions combine between the technical and human aspects of protecting critical infrastructure systems.

Then we choose dataset that helps us to do the machine learning evaluation, so we decided to use Kaggle to collect a synthetic dataset which was "Global Cybersecurity Threats (2015 - 2024)", as this dataset was chosen because it covers a 9 years of cyberattacks and threats which is a good long period, covers multiple countries and critical infrastructure systems like banking and retail, and include attacks that are related to critical infrastructure systems. We chose a synthetic dataset to prove that synthetic datasets are not always suitable for machine learning evaluation, and because we are working within an academic aspect, this synthetic dataset was suitable for our study. The

use of Kaggle as a data collection method was because Kaggle is free to use and has many accessible datasets that are crucial for research purposes.

Then we explained our onion model, as we chose critical realism for philosophy because we needed a philosophy that supports both technical analysis and human points of view. For the theory development approach, we decided to go for an inductive approach because we are not testing an existing theory, but conclusions are explained from the data. For the methodological decision, we decided to go for mono-method quantitative because both research questions rely on numerical analysis and we thought that there is no need for a qualitative analysis. For the time horizon and strategy, we decided to use survey and experiment because we are doing an experiment for secondary data, and questionnaire for primary data, also we chose cross-sectional time horizon because all data were collected and analysed at one point.

Then we chose the techniques and procedures used, as we chose Google Forms to create the survey and python for the experimental analysis. We used Google Forms because it's simple and easy to create questionnaire with, I can create a questionnaire very fast and responses can be exported in different formats, and google forms helps me to make an analysis as it shows different type of charts which makes it easier to measure awareness, training, and insiders affection on critical infrastructure systems. We also used python for data preprocessing model training, and evaluation because python includes a wide range of libraries that help us do the whole modelling with.

Then we made preprocessing, and the decision to do preprocessing was very important because preprocessing made sure that the dataset is ready for machine learning modelling, as we checked for missing or null values, doing label encoding, standardization for numerical features to make sure that models don't think higher number is better, and feature separation.

Then we chose five machine learning models rather than using only one model to view the performance across multiple models which gives us an accurate results, so, we selected five models (Random Forest, SVM, XGBoost, K-Nearest Neighbors, and Naïve Bayes) because these five models show a different learning ways through trees, distance, probability, or margins which made sure that the findings were fully correct. So, we split the dataset into 80% training, and 20% testing because it is a common practice in machine learning evaluation. Also, we decided to select four evaluation metrics (Accuracy, F1-Score, Recall, and precision) because sometimes accuracy alone is not enough to measure the performance.

The survey questions chosen were a closed-ended question with a scale from 1 to 5 to help us do a quantitative analysis and improve clarity in the answers, as the question was designed to measure the training and insider threats effects on critical infrastructure systems, as each question contributed to answering the research question.

We have been through many challenges in this research that delayed the research process in different parts of the project, but we were able to solve these challenges effectively which allowed us to complete the research and meet the research objectives.

At first, we had a challenge related to how to write and start the research, and structuring the research, as we wrote multiple false or wrong introductions which delayed the process at an early stage. But we did not give up, so we looked at different existing academic research papers related to cybersecurity and critical infrastructure systems to understand how they introduce readers and how research papers are structured including introduction, gaps, goals, objectives, research questions, etc.

Also, the study at first was too general and focused on the overall protection, but Dr Fayez mentioned that the study should be focused on a specific aspect, so we addressed this challenge by focusing the research on two aspects (AI attack prediction, and Human factors impact).

Another challenge was we had some difficulties in finding reliable research studies that focus on critical infrastructure systems protection, as most of the studies were too general or too focused on a single part which took us a lot of time to find what we are looking for, so, we looked at a lot of researches to build a good backbone for us and for this study.

Also, there was a challenge to the collection of survey data, as the number of the responses at first was very low and we need a larger number of responses to improve the reliability, so, we addressed this challenge by not only send it to IT students at Al Hussein Technical University, but also, we sent the survey to IT students from multiple universities and did not only limited it to a single university, and we sent the survey multiple times to remind the audience who did not fill the survey to fill it, which helped in increasing the number of the responses.

There is a challenge related to the dataset selection, as it took us a lot of time to find a suitable dataset that is related to critical infrastructure systems and support machine learning modelling, as many dataset were too general and not related to critical infrastructure systems, so, we addressed this challenge by looking at a lot of dataset and try to find the most suitable one and related to critical infrastructure systems.

Another challenge was choosing the appropriate tools for making the survey, making the data preprocessing step, and making the modelling step, so we looked at many tools and realized that Google Forms is the suitable one for making the survey, and Python is the suitable one to make the data preprocessing and modelling step.

The last major challenge was during the experimental phase, as there were some errors in Python preprocessing and modelling code which caused some delays, so, we addressed this issue through debugging and problem solving, and after finding the error it was a very simple error related to the parentheses.

There was a challenge related to time management, as it was intended in each week we write and solve one part of the research, but due to the challenges mentioned earlier, there were delays, so, we addressed this challenge by rewriting a timeline for completing the research.

Before this research, I had a really minor knowledge in research papers, so while writing this research I was able to improve my research skills including reading research papers, understanding research components, understanding how research papers are created, and it also improved my knowledge in cyber security and especially in the protection of critical infrastructure systems. The experience helped me to gain and improve knowledge in searching for research papers or datasets, thinking more to produce effective thinking of the decisions and ideas, choosing a field that I'm interested in that needs more studies. I learned more about onion model, machine learning skills through doing preprocessing and modeling with python, machine learning models because before I did not know much about these models but now, I have gained a deeper understanding and knowledge of them. Also, during this research I built a strong knowledge about the technical and human factors that affect the protection of different systems, and especially critical infrastructure systems, and gained knowledge on data quality influence in machine learning modelling. As a result, I gained more confidence to work alone in the future on research papers, so, future research

will be conducted to practice and publish research papers and they will be better than this paper as now I know the process of writing a research paper, I know the components of research paper, and I know more about machine learning models.

If I had the ability to repeat this research again, the research process would follow a new process as shown in figure 5, as we developed this process based on the challenges we faced throughout the project.

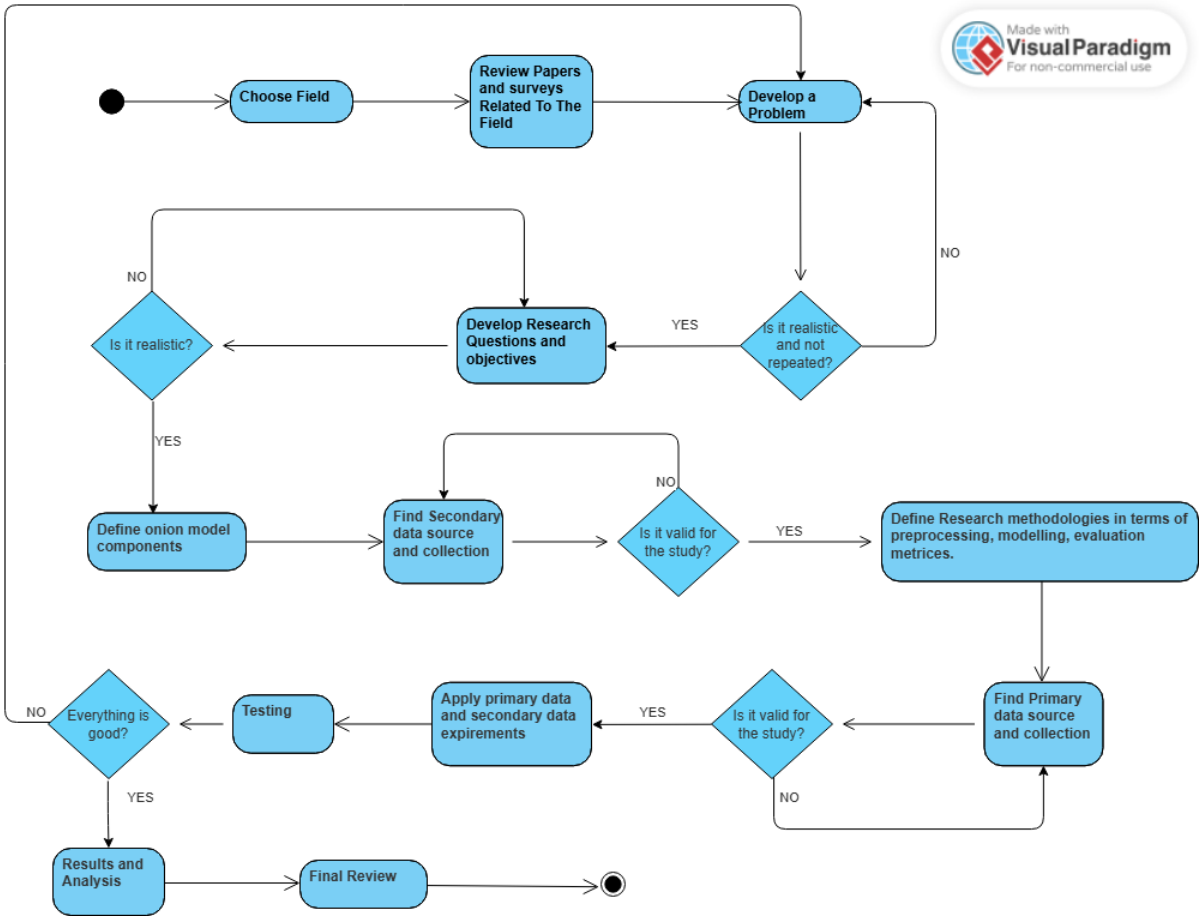


Figure 6: Activity Diagram for Revised Research Process

At the new version of the research process, we will start by choosing a field of our interest, followed exactly by reviewing existing papers and surveys related to the chosen field which is very important to produce a high quality problem statement, as by that we can identify the gaps in existing research papers, and focus on not choosing a problem that is repeated.

After that we will define and develop a problem which will be our specific field of study because the placement of that activity right after reviewing the papers would give a more realistic process because we don't want to make a research paper that is repeated and want to produce something new, right after developing the problem there is a decision block to view, and test if the produced problem is realistic and not repeated, if no we will go back to developing another problem, if yes we will go to the next activity.

The next activity is about developing research questions and research objectives based on the field, existing papers, and the chosen developed problem which will make sure that the research questions and research objectives can produce something new to the readers and people of interest. Right after developing research questions and objectives we will validate if these questions and objectives are realistic, if no, we go back to developing other research questions and objectives, if yes, we can go to the next activity.

The next activity is about defining onion model components, and it was placed here to explain to the readers the components and help us understand how we will solve the research question which would save time while solving the research questions and meeting the research objectives.

Then we will find secondary data source and collect these data, and it is after the onion model activity which will help us find a reliable and good source to collect these data, and there is a validation to confirm if the dataset supports the research question and is valid for the study, if no, the process go back to select a dataset, if yes, the process proceed to the next activity.

The next activity is defining the research methodology in terms of data preprocessing, modelling and evaluation metrics, and this activity was added here so when we apply them, everything will be clear and documented.

Then we will find primary data source and collect these data based on the onion model we provided earlier, so then we will validate if these data are valid for the study, if no, the process go back to developing and collecting primary data, if yes, the process proceed to the next activity.

The next activity is applying experiments on primary and secondary data in parallel, and then testing to see if there is something went wrong, if everything is good we proceed to the next step, if no, we go back to the beginning at the problem development activity to check what went wrong.

If everything is good, we analyze results and present it, then followed by a final review.

This approach is working by validation-based approach, as it was the main difference between our research process and the revised research process, and it was like that because in the old research process we kept choosing things that appear they are wrong later, but with validation right after some activities, we can find issues early. Also, after testing phase, if things were not good, we go back to the start to see what went wrong instead of changing the whole study or problem.

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